Summery regarding Shift-Registers

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What's a Shift Register

A shift register is a sequential digital circuit built from a series of flip-flops, each storing one bit of data. On each clock pulse, the content of each flip-flop shifts to the next flip-flop in the chain. This allows binary data to be stored temporarily and moved or manipulated serially or in parallel.

Shift registers are widely used in digital electronics for data storage, data transfer, serial-to-parallel or parallel-to-serial conversion, time-delay elements, and buffering in communication and control systems.

Depending on how data is loaded into (input) and read out of (output) the register – serially (bit by bit) or in parallel (all bits at once) – there are four main types of shift registers.

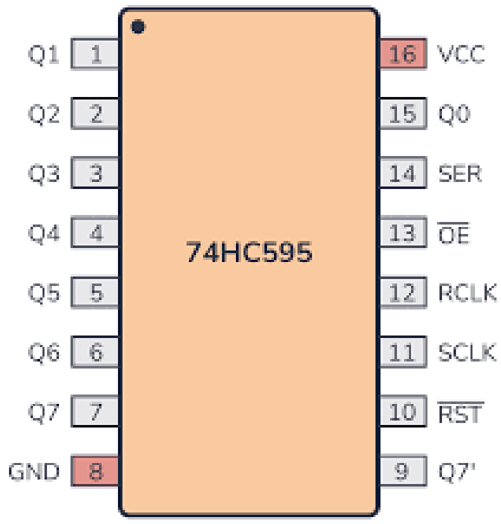


Diagram for the 74HC595 Shift Register

Types of Shift Register

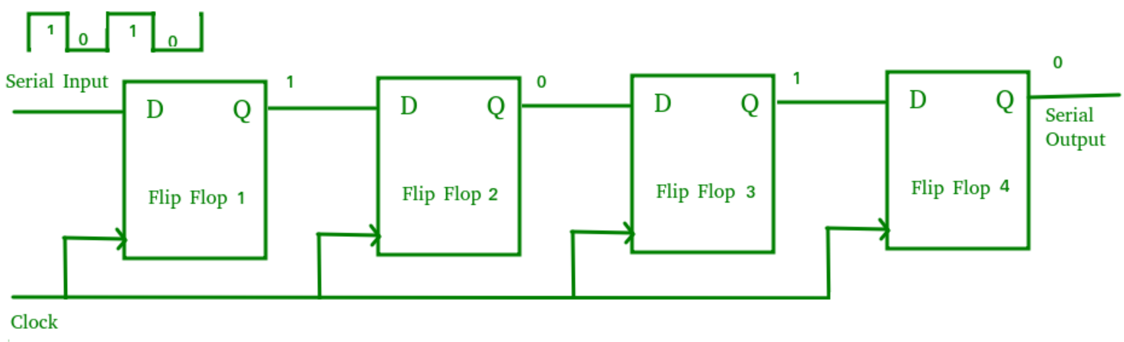
1. SISO
2. SIPO
3. PISO
4. PIPO

1. SISO(Serial in, Serial out)

   A Serial-In Serial-Out (SISO) shift register accepts data one bit at a time through a single input line and outputs data serially, using a series of flip-flops all triggered by the same clock. The main use of a SISO is to act as a delay element

   Data enters and leaves one bit at a time in a serial sequence

   Typically made of multiple D flip-flops connected in series and synchronised by a common clock



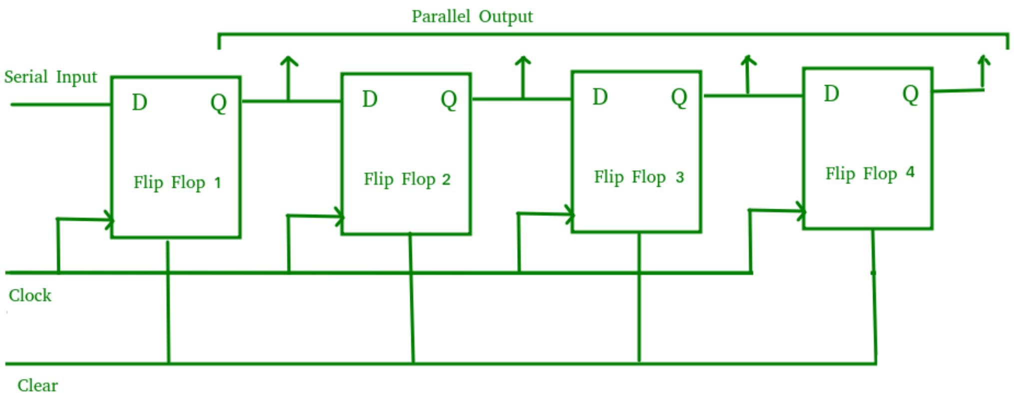
Circuit diagram for SISO Shift Register

2. SIPO(Serial in, Parallel out)

A Serial-In Parallel-Out (SIPO) shift register accepts data serially through a single input line and produces all bits simultaneously as a parallel output, using multiple synchronized flip-flops. The main use of the SIPO register is to convert serial data into parallel data.

Data is input one bit at a time and outputs all stored bits in parallel.

Consists of connected D flip-flops with a common clock and a clear (CLR) signal to reset all flip-flops



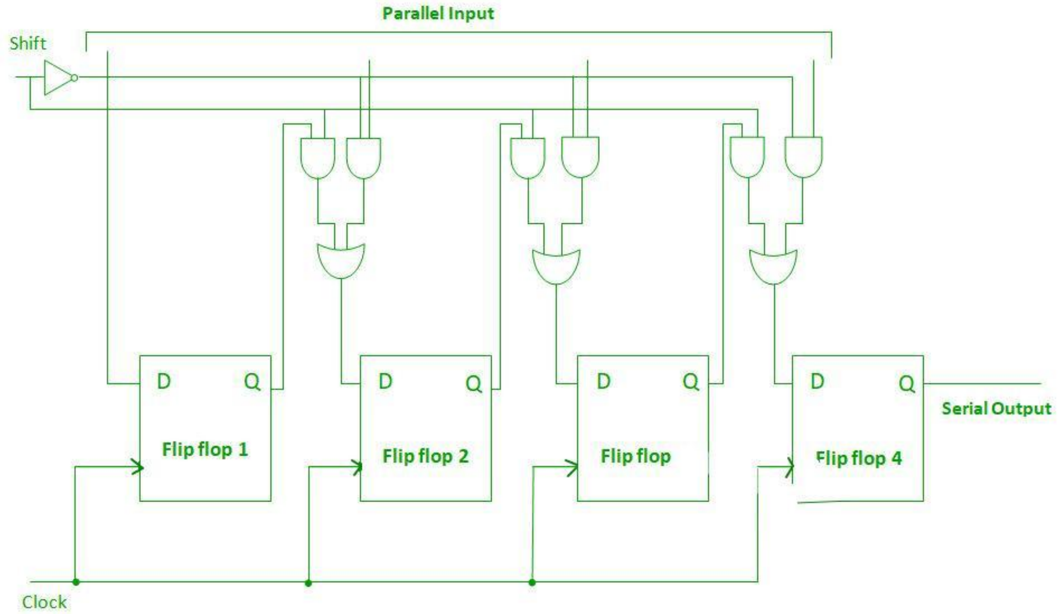
Circuit diagram for SIPO Shift Register

3. PISO(Parallel in, Serial out)

A Parallel-In Serial-Out (PISO) shift register accepts parallel data simultaneously into each flip-flop and shifts it out serially through a single output line using synchronized flip-flops and multiplexers. It is used to convert parallel data to serial data

Data is loaded in parallel and then shifted out one bit at a time serially

Consists of D flip-flops with inputs controlled by multiplexers selecting between parallel data and previous flip-flop output, all clocked together



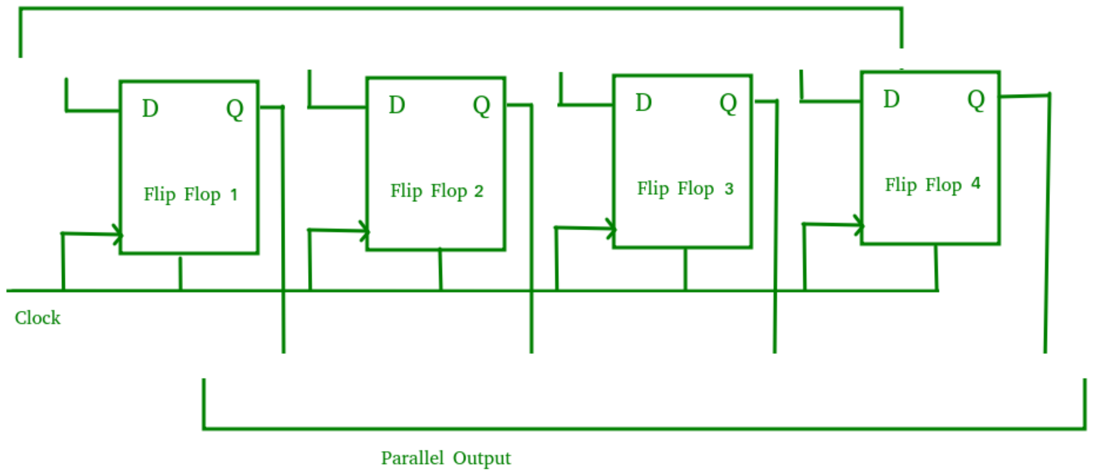
Circuit diagram for PISO Shift Register

4. PIPO(Parallel in, Parallel out)

A Parallel-In Parallel-Out (PIPO) shift register accepts and outputs data simultaneously across all flip-flops without any serial shifting, using synchronized control signals. It is used as a temporary storage device and also acts as a delay element.

Data is loaded and retrieved in parallel from each flip-flop at the same time

Includes D flip-flops with no interconnections, controlled by common clock and clear (CLR) signals



Circuit diagram for PIPO Shift Register

Applications of Shift Registers

1. Serial-to-Parallel Conversion – e.g. receiving serial data (from a serial communication line) and converting it to parallel form for processing. SIPO registers are ideal for this.

2. Parallel-to-Serial Conversion – e.g. sending data over a single data line by serializing parallel data from a microcontroller or parallel bus. PISO registers are used here.

3. Delay Elements / Data Timing – e.g. using SISO registers as delay lines to delay digital signals by a specified number of clock cycles.

4. Temporary Data Storage / Buffering – e.g. PIPO registers to hold multi-bit data and retrieve all bits at once.

5. Multiplexing & Demultiplexing: Converting data between serial and parallel forms to suit different parts of a digital system (e.g. microprocessor ↔ peripheral interface).